

2024 Presidential Election

IS ABORTION A

POLITICAL MATTER?

Predictive Modeling Case Study by
Harini Karthikeyan

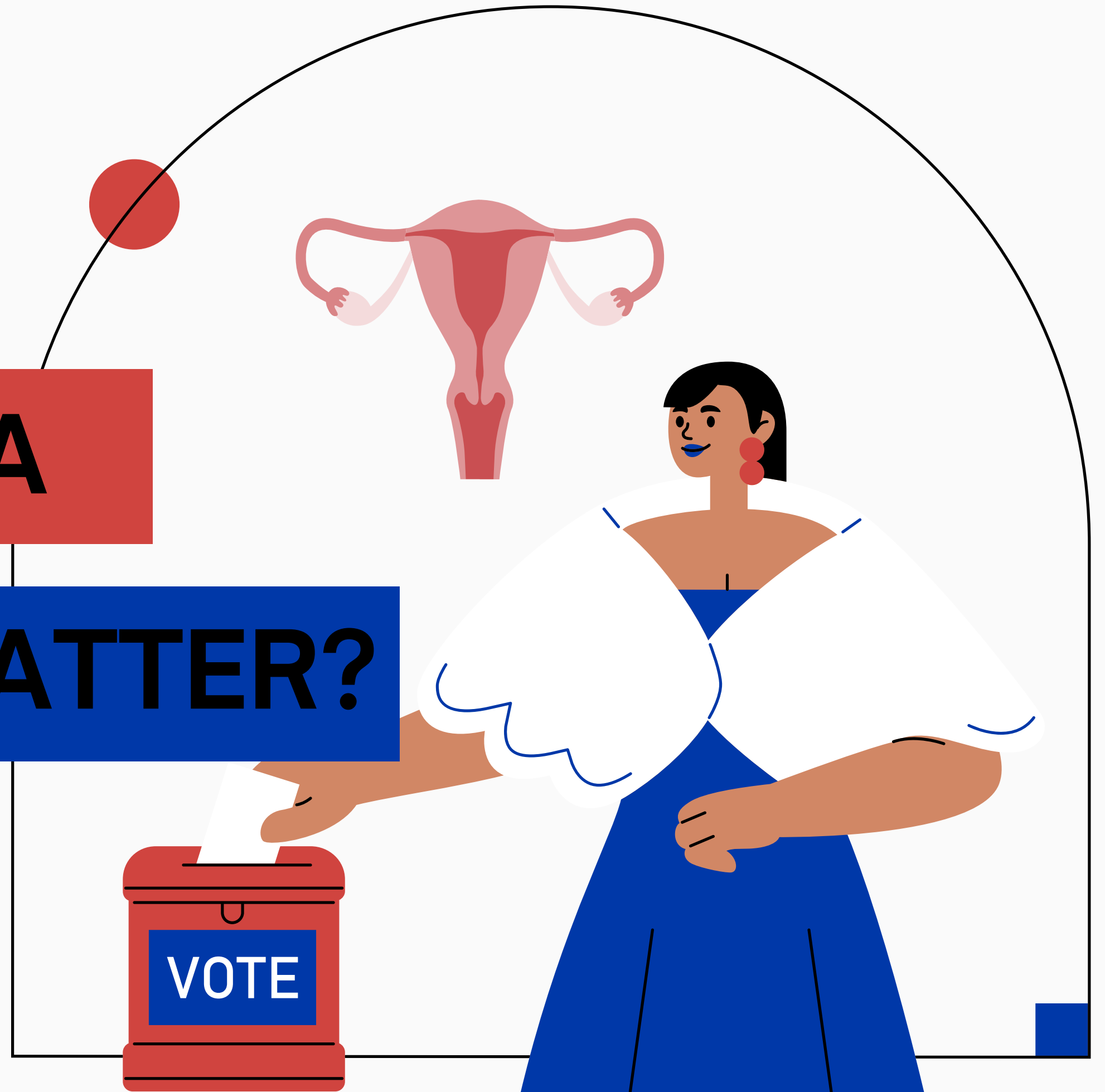
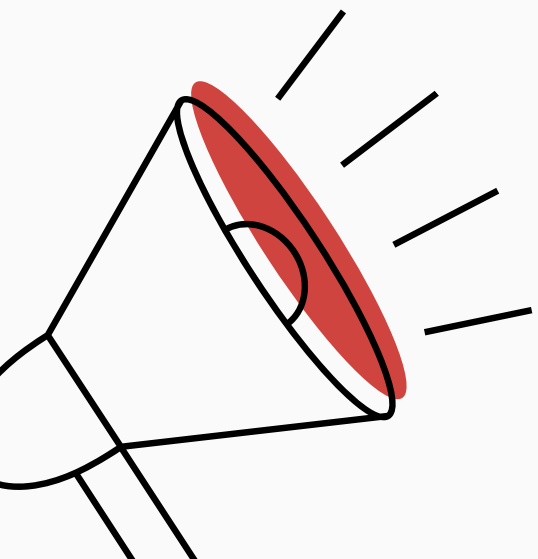


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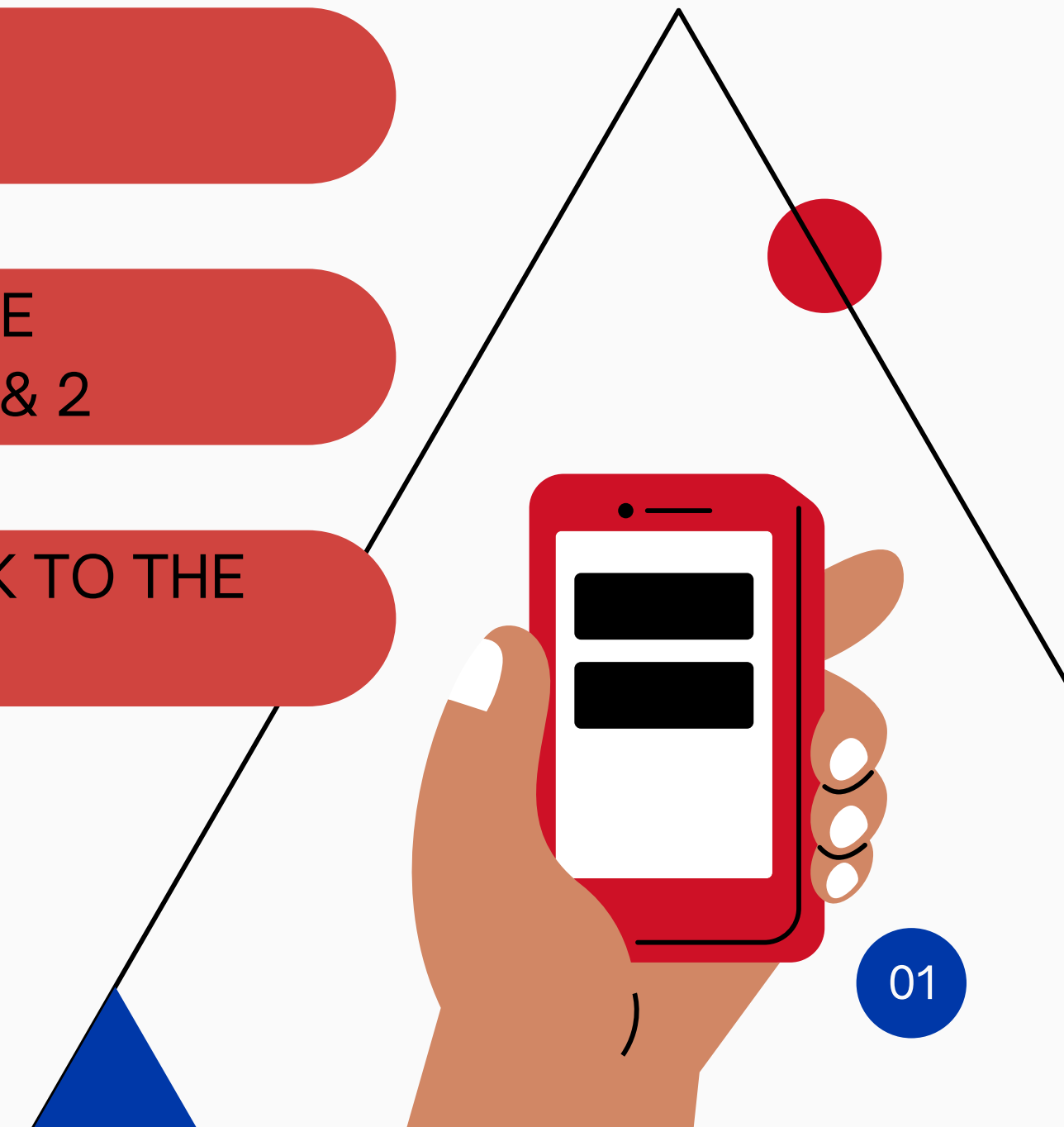
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METHOD TO DISTINGUISH STATES BY ABORTION LAWS

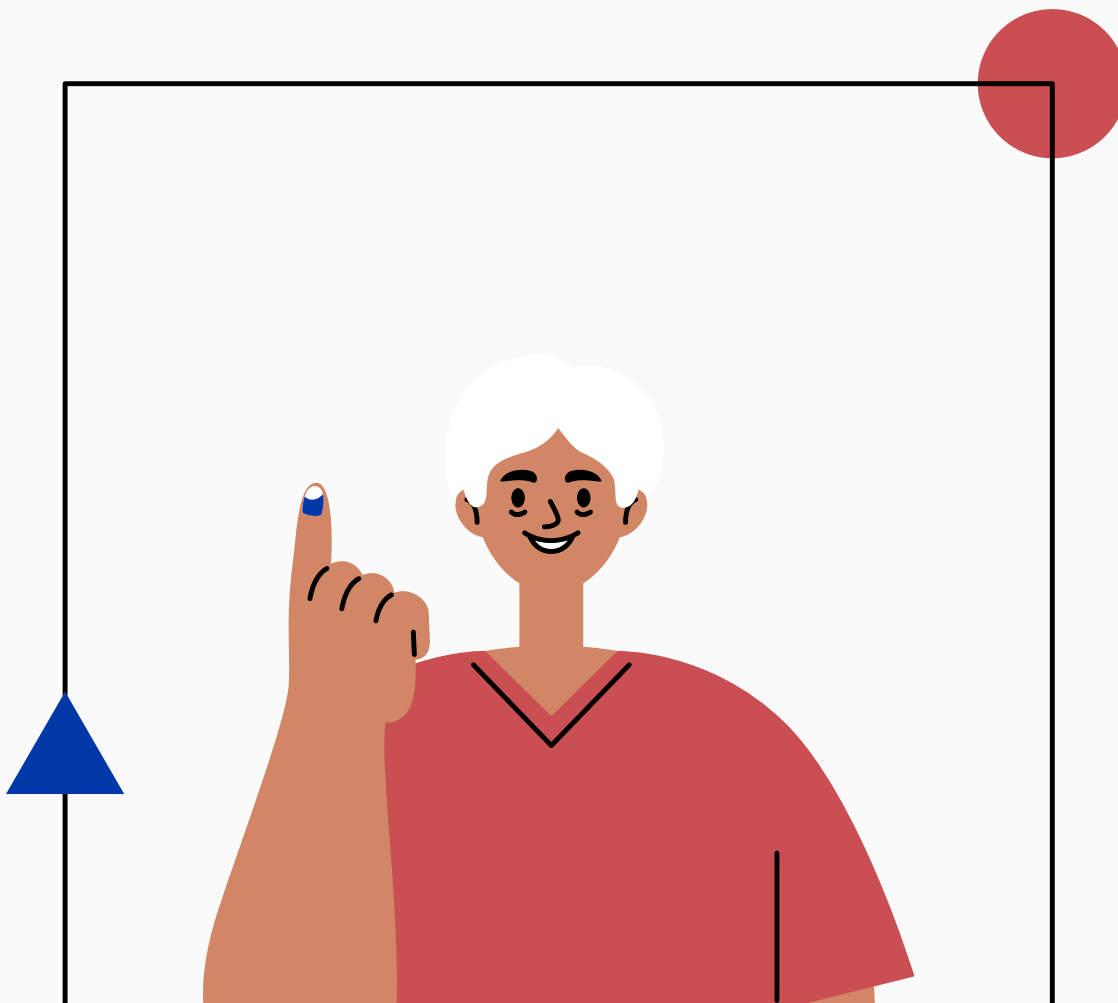
- To evaluate voting behavior across different reproductive healthcare policy environments, I categorized states based on their abortion laws in effect during the 2024 presidential election.
- Using state statutory frameworks and legal documentation, states were classified into two primary categories: Abortion Legal/Protected and Abortion Illegal/Banned.
- Using the tidyverse library in R, I created a new categorical variable (abortion_status) using the mutate() function to assign each county to its respective state policy group.
- States enforcing partial or gestational-age bans were assigned NA values and subsequently removed using the filter() function.
- This yielded a clean dataset consisting strictly of counties operating under total ban or fully protected policy regimes, ready for statistical hypothesis testing.



HYPOTHESIS TESTS

Democratic Votes Casted vs Abortion Legality

Null Hypothesis : There is no difference in mean Democratic vote percentage between counties in states where abortion is illegal versus legal.
Alternative Hypothesis : There is a statistically significant difference in the mean Democratic vote percentage between counties in states where abortion is illegal and counties in states where abortion is legal.



Republican Votes Casted vs Abortion Legality

Null Hypothesis : There is no difference in mean Republican vote percentage between counties in states where abortion is illegal versus legal.
Alternative Hypothesis : There is a statistically significant difference in the mean Republican vote percentage between counties in states where abortion is illegal and counties in states where abortion is legal.

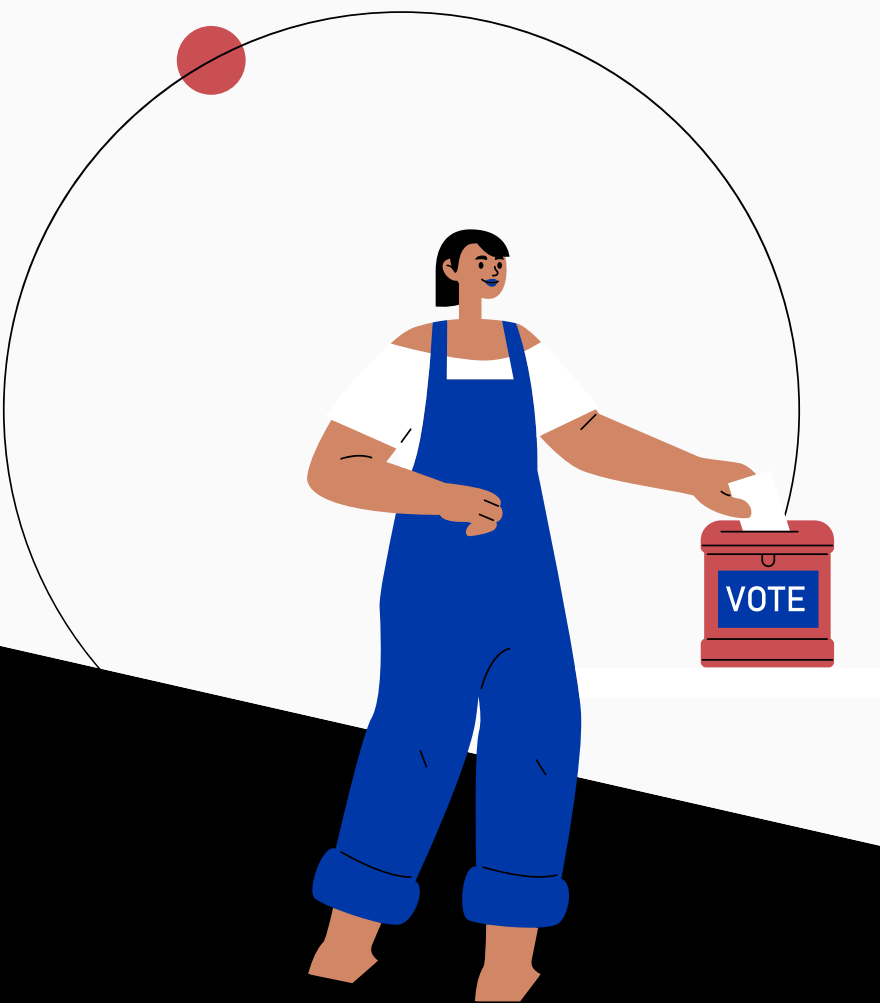
RESULTS

Democratic Hypothesis

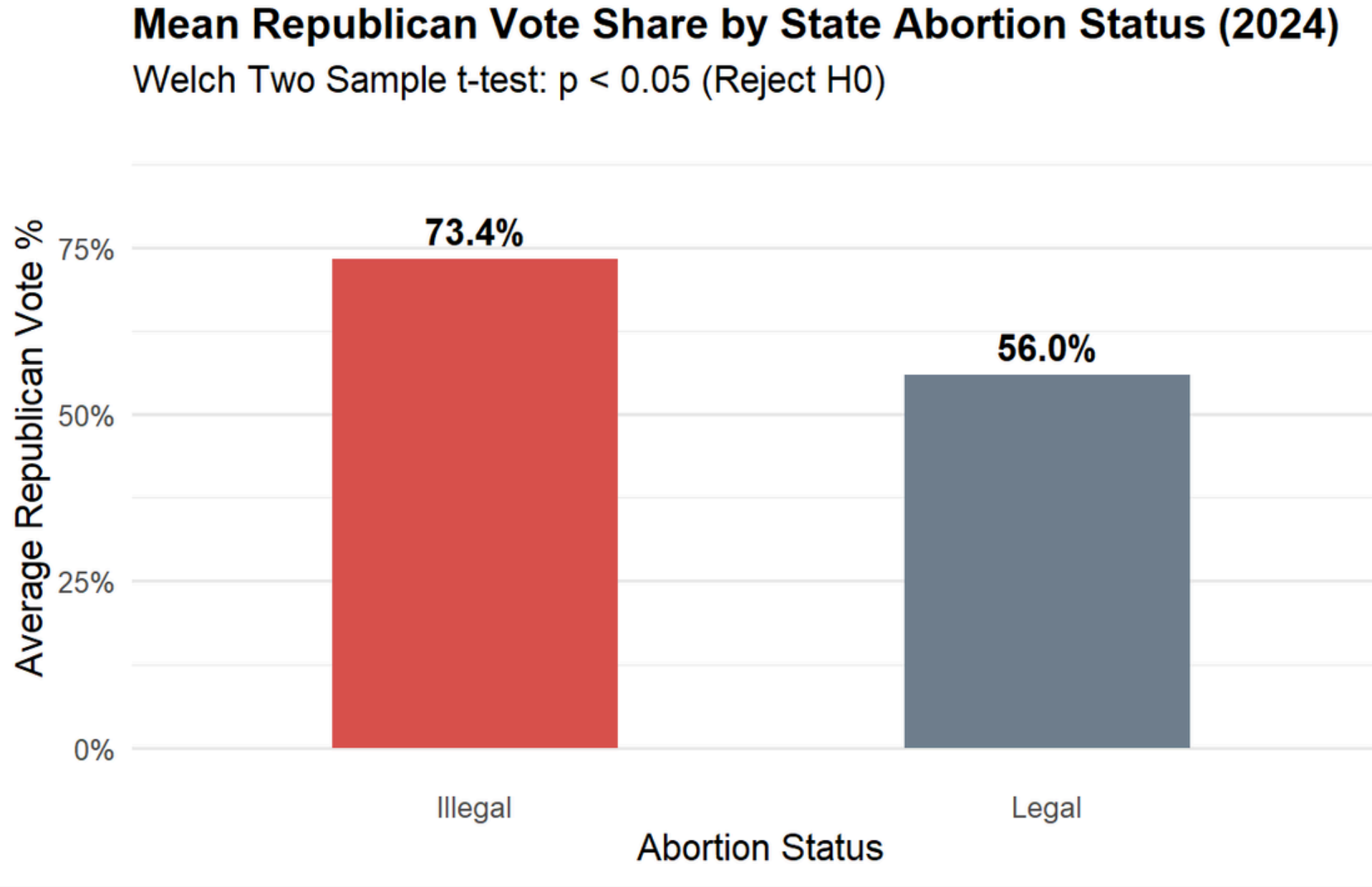
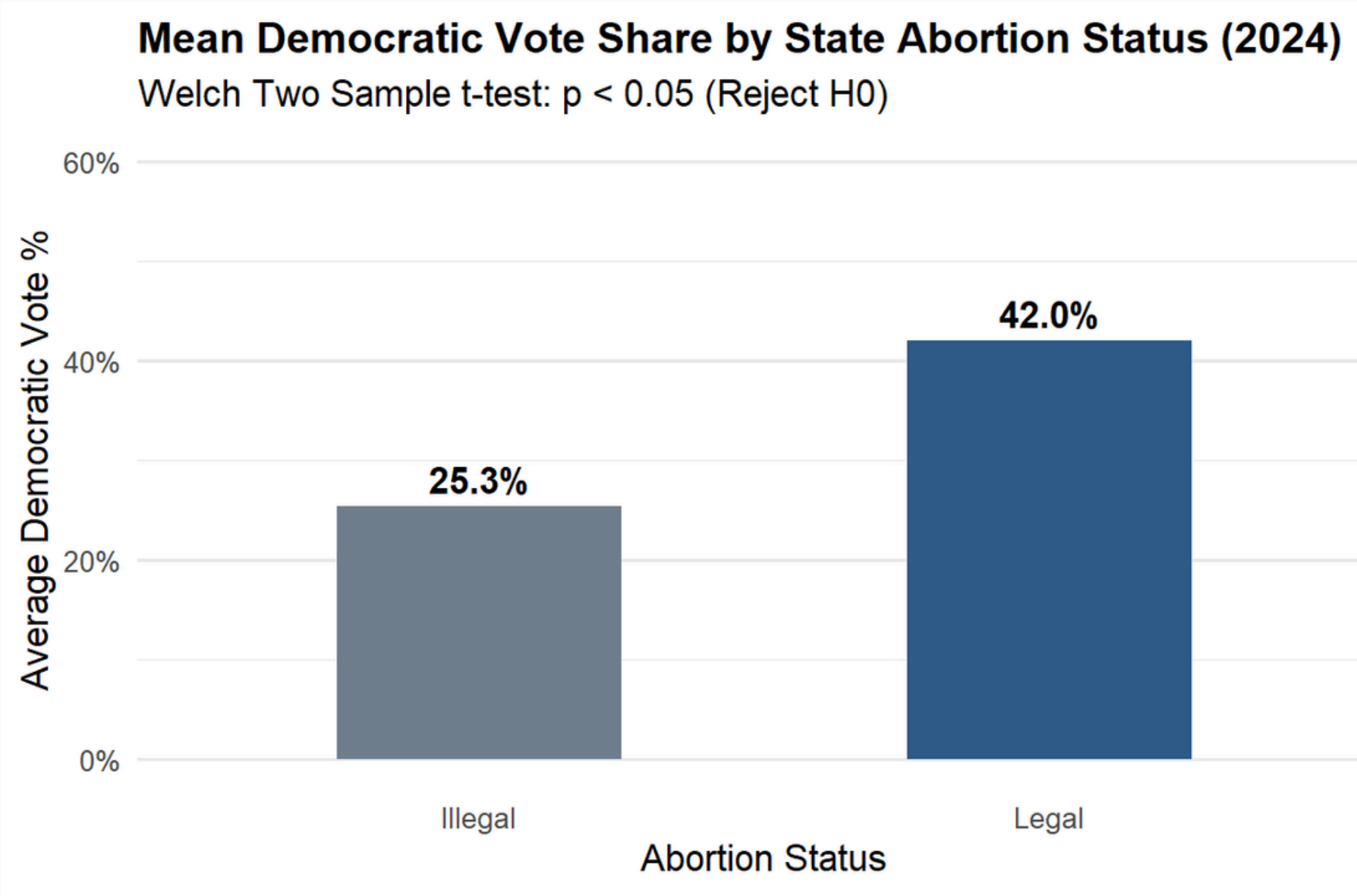
- Because the computed p-value is less than the standard threshold ($2.2 \times 10^{-16} < 0.05$), there is statistically sufficient evidence to warrant the rejection of the null hypothesis.
- What does this mean? There **IS** a difference in the mean Democratic vote percentage between counties in states where abortion is illegal and counties in states where abortion is legal.

Republic Hypothesis

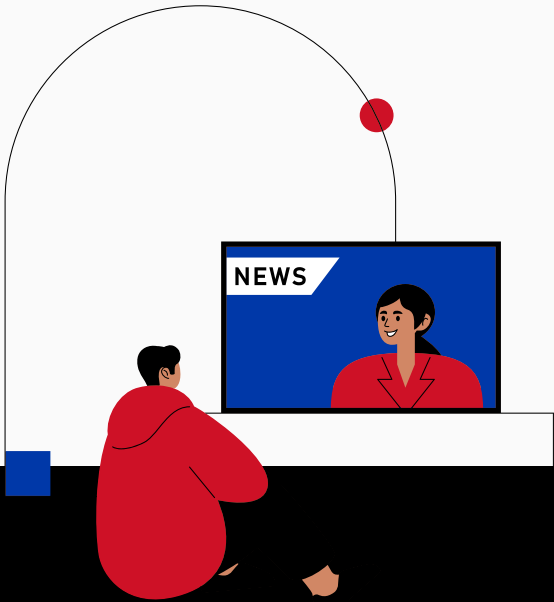
- Because the computed p-value is less than the standard threshold ($2.2 \times 10^{-16} < 0.05$), there is statistically sufficient evidence to warrant the rejection of the null hypothesis.
- What does this mean? There **IS** a difference in the mean Republican vote percentage between counties in states where abortion is illegal and counties in states where abortion is legal.



GRAPHS - BAR PLOTS FOR EACH HYPOTHESIS

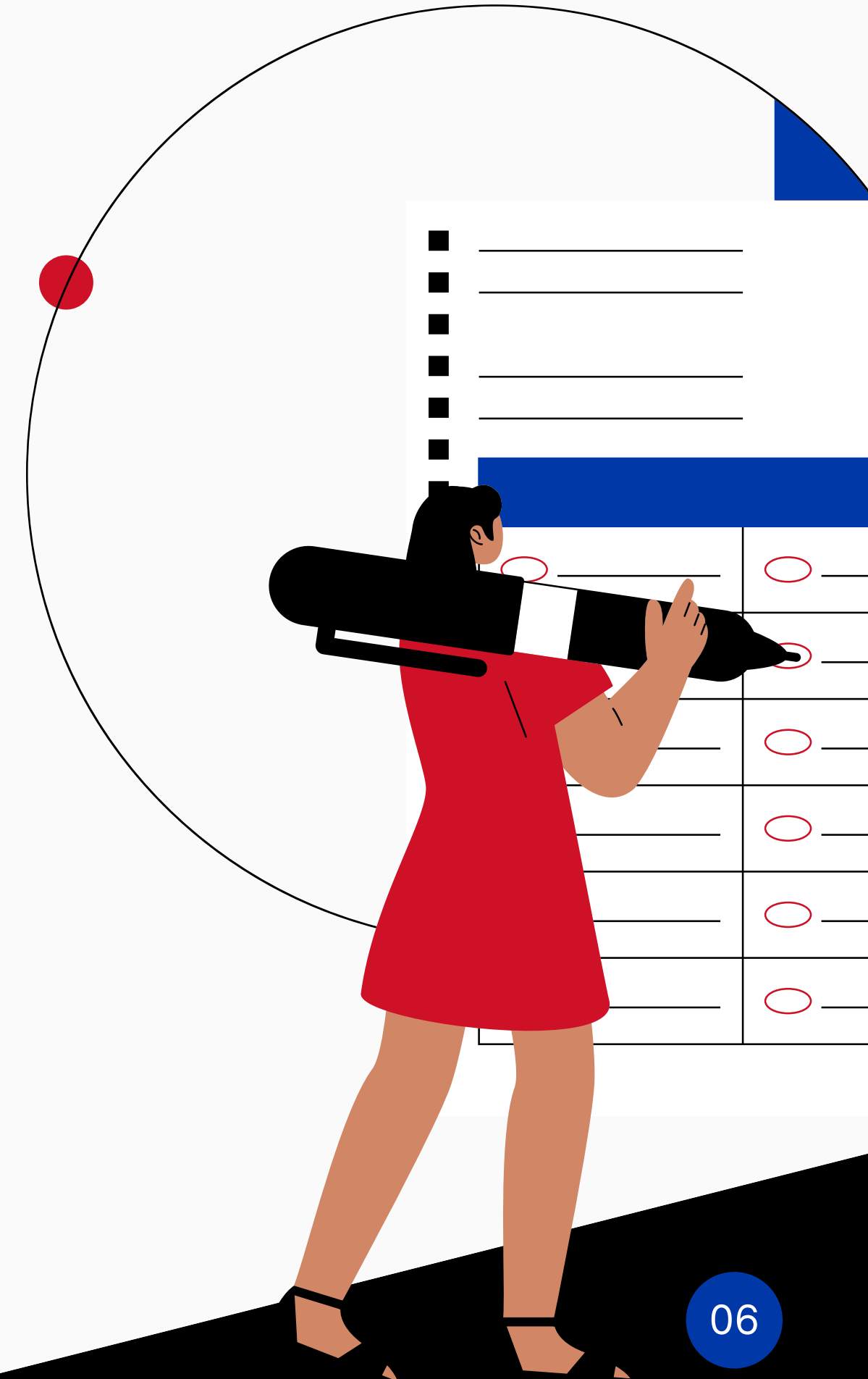


Note all graphs and results were produced using 'R' statistical programming language



INVESTIGATING THE DIFFERENCE PT 1

- As proven by both Independent t-Tests, we know that there are guaranteed differences in both respective political parties' mean vote percentages between counties in states where abortion is illegal and counties in states where abortion is legal.
- But, what is the difference? Is one group greater than the other?
- From the graphs on the previous slides, here is what we can observe :
- The mean of the **Democratic** share of votes is higher in states where abortion is legal than states where abortion is illegal.
- The mean of the **Republican** share of votes is higher in states where abortion is illegal than states where abortion is legal.
- These are statistically solid results but now the question is “What is the reason behind this difference?”



INVESTIGATING THE DIFFERENCE PT 2

1. Political Party Alignment (The Main Driver)

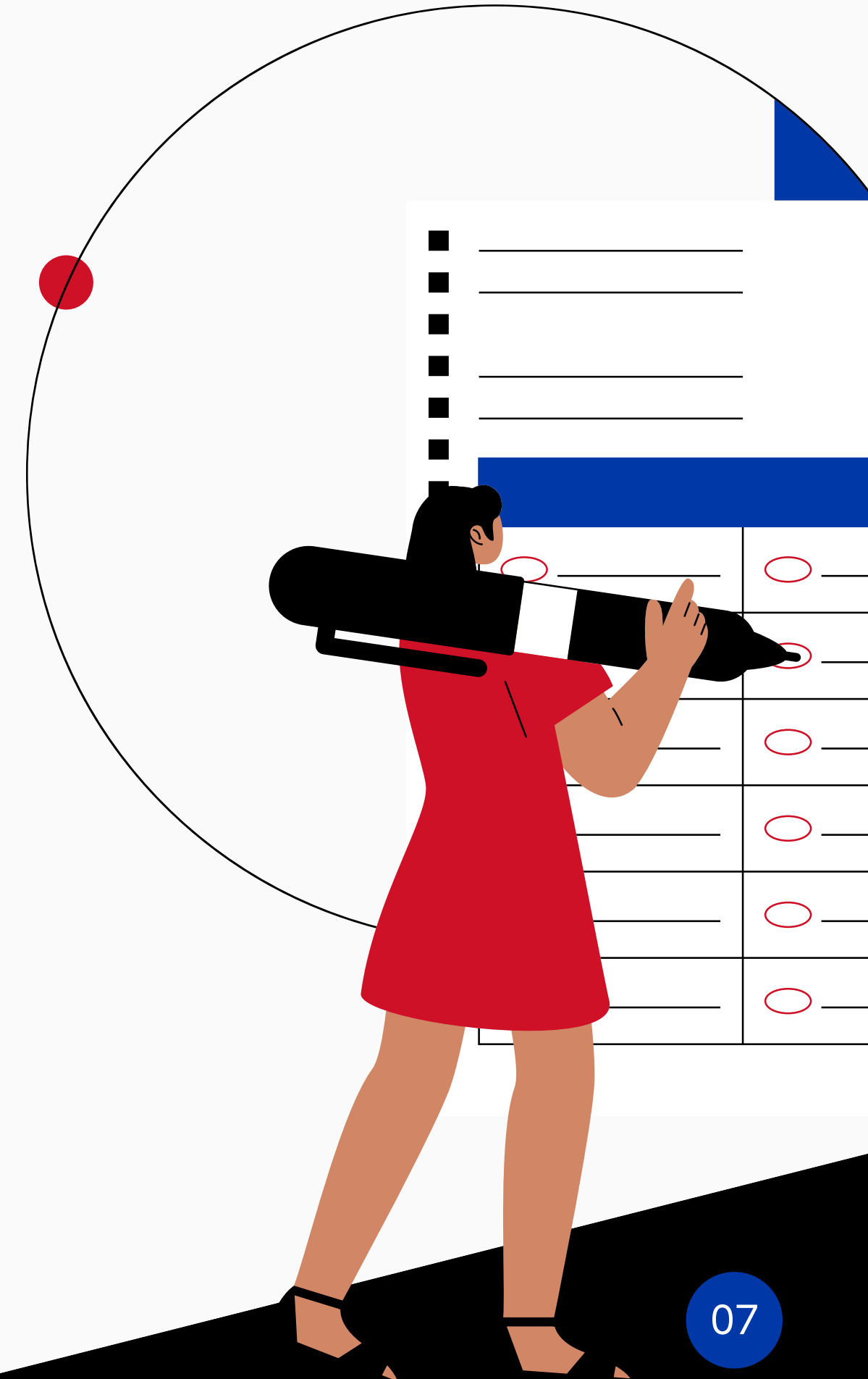
- States pass laws based on which party is in power:
- Republican States: Elect conservative leaders who pass abortion bans and vote for Republican presidential candidates.
- Democratic States: Elect liberal leaders who protect abortion access and vote for Democratic presidential candidates.

2. The Urban vs. Rural Divide

- Geography plays a major role in political voting patterns:
- Urban/Suburban Counties: Tend to have higher proportions of college-educated voters and lean Democratic. States with large city hubs (e.g., California, New York) tend to protect reproductive rights.
- Rural Counties: Tend to lean heavily Republican (often 70–80% vote share). States with predominantly rural county distributions (e.g., Texas, Mississippi, Oklahoma) tend to enact complete abortion bans.

3. Relevance of Reproductive Rights Post-Dobbs (2022)

- Since the 2022 Supreme Court ruling, abortion has heavily driven voter turnout:
- Democratic States: Motivates voters to protect and expand abortion access.
- Republican States: Aligns with conservative voters who support complete bans.



FULL CIRCLE : BACK TO THE TITLE QUESTION



The Direct Answer: Unmistakably, Yes.

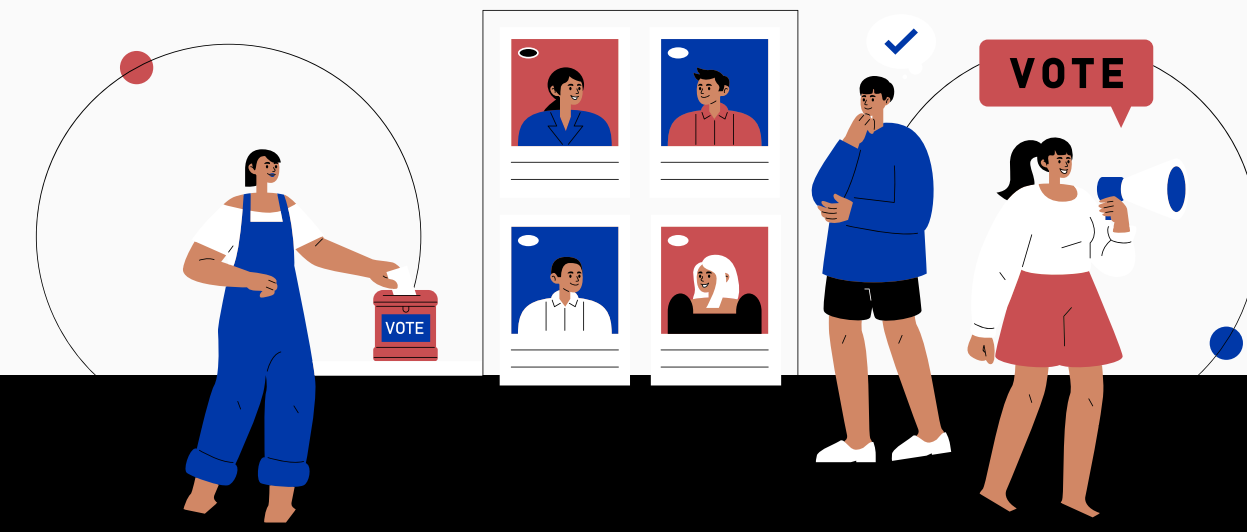
- Our data proves that state abortion policies and national presidential voting patterns are deeply rooted in modern American politics.

The Statistical Evidence

- **Democratic Vote Share:** Counties in states where abortion is legal voted significantly more Democratic (42.0%) than counties in states where it is restricted (25.4%).
- **Republican Vote Share:** Counties in states with abortion bans voted overwhelmingly more Republican (73.4%) than counties in states where abortion is protected (56.0%).

Why This Matters (The "Full Circle" Takeaway)

- Abortion is no longer just a healthcare or personal debate.
- It is now a defining line in American party politics:
- State Policy Reflects Presidential Choice: The same conservative voters who elect state lawmakers to ban abortion are driving high Republican presidential vote shares.
- Reproductive Rights Drive Voter Mobilization: Since 2022, abortion policy has become a core issue that defines party identity and motivates turnout on both sides.



THANK YOU!
Questions? Email
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